

A student team building an autonomous boat from the ground up.



HOW WE MODEL IT

First-principles mixed-integer linear programming

We build the model from physics up. Each constraint cuts the space of possible boats, and the optimizer finds the fastest one left. Hull resistance comes from **Savitsky's planing-hull model**. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

ENGINEERS FROM ACROSS UW-MADISON

Experience at Tesla, Northrop Grumman, Xcel Energy and Milwaukee Tool

Industrial

Optimization, composites and manufacturing

Mechanical

Structures, CAD and fabrication

Electrical

Power, battery and high voltage

Computer

Embedded, firmware and software

OUR APPROACH

Hull

Plane well and stay light.

- Hybrid planing composite hull
- Shape tuned by optimization
- Resistance checked in CFD first

Electric

Every watt-hour to speed.

- High-voltage battery storage
- Propulsion sized by the model
- Propulsion and cooling co-designed

Autonomous

Manage drift and controls.

- GPS, IMU and LIDAR sensing
- Firmware protects the battery
- Software plans the route

TIMELINE

Oct 25	Current hull design frozen	Jan 22	First float and tow test The hull floats with deadweight for the real weight, and is towed behind a motorized boat to validate the hull design and quality.
Nov 15	Preliminary design review Full CAD and circuit design, plus a rough proof of concept, even a cardboard boat with boxes for the components, to visually check dimensions and catch issues. Critical and long-lead components are ordered.	Jan 29	Zeroth nautical mile The first time everything is powered on in the hull: all microcontrollers, sensors and actuators. The first motor and propeller spin with the boat lifted in the air, and the first propulsion and navigation test commanded by the autonomous pipeline.
Dec 1	Mechanical and electrical design frozen With the full BOM and the order.	Feb 1-14	Slow drives on the water Fully manual first, by remote command, then adding autonomy over time.
Dec 1-31	Build the hull and composite parts Plus any custom-milled parts.	Feb 15	Start increasing speed
Jan 11-28	Manufacture and assemble everything All mechanical and electrical components, with dry fits, electrical assembly out of the boat first, and debugging.	From Feb 15	Test as much as possible Right up to the competition, working through the hardware and software issues that come up.
Jan 11-Feb 26	Build likely spare parts, in parallel Spares for the parts most likely to be used.		

Sponsor the build

We are looking for funds, parts, advisors and people to build our BOM as we design.

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CONTACT

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